

Zhihao Wang

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EDUCATION

Carnegie Mellon University, School of Computer Science <i>Master of Software Engineering</i>	Pittsburgh, PA Aug 2021 – Dec 2022
China University of Mining and Technology, Beijing, School of Science <i>Bachelor of Science in Information and Computing Science</i>	Beijing, China Sep 2013 – Jun 2017
Peking University, National School of Development <i>Bachelor of Economics (double major)</i>	Beijing, China Sep 2014 – Jul 2017

EXPERIENCE

Temporal Technologies <i>Staff Software Engineer, Infrastructure Foundations</i>	Pittsburgh, PA Sep 2026 – Present
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- Improving the scalability and reliability of Temporal Cloud core infrastructure.

Duolingo <i>Staff Software Engineer, Cloud Operations</i>	Pittsburgh, PA Apr 2026 – Sep 2026
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- Served as technical lead for a Temporal infrastructure project team of 5 engineers: defining project direction, mentoring engineers on Temporal internals and infrastructure patterns, and partnering with teams across the org to design and onboard new use cases, as the platform grew to 15M+ Temporal Cloud actions/month, reinforcing Temporal as core infrastructure at Duolingo.
- Shipped KEDA-based task-queue autoscaling and Kubernetes-native versioned deployments for Temporal workers, after leading a fully transparent migration of the entire worker fleet from ECS to EKS across 20+ namespaces with no action required from workflow owners; also rolled out a payload codec solution giving teams field-level control over sensitive workflow data org-wide.
- Executed zero-downtime cross-account and cross-region migrations of production Aurora PostgreSQL, DynamoDB, and other datastores (S3, cache, etc.) from a monolithic AWS account to dedicated accounts for PII/UGC isolation and scalable multi-account management; leveraged Terragrunt to abstract account-level infrastructure (VPC, subnets, peering, etc.); automated CIDR planning and programmatically generated migration playbooks to de-risk each account cutover.

<i>Senior Software Engineer, Cloud Operations</i>	Feb 2023 – Mar 2026
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- Championed Temporal adoption at Duolingo from 0→1: introduced the technology, built foundational infrastructure from scratch, and scaled it into an org-wide durable workflow platform powering 160+ production workflows across 20+ teams — establishing Duolingo as a recognized Temporal reference implementation featured in Temporal’s own case studies and conference talks.
- Built the Temporal Self-Service Platform with Temporal Nexus, enabling 15+ CloudOps self-service workflows and saving thousands of engineering hours annually; integrated Claude Agent SDK, OpenAI Agents SDK, and Codex CLI to power Temporal-based agentic workflows for multi-step AI-driven automation.
- Improved developer experience in the Temporal monorepo — refactored tooling and deployment scripts, shipped Agent Skills for workflow authoring, and reduced CI, build, and pre-commit times by 50%, accelerating onboarding for new teams.
- Stepped into de facto DBRE ownership: built production-grade Aurora PostgreSQL monitoring from scratch, implemented full data store disaster recovery via Temporal workflows, and upgraded 30+ Aurora clusters v11→v14 in under one month with zero downtime and zero learner impact, backed by custom Grafana dashboards for table size, slow queries, and blocked queries.

data.ai (formerly App Annie) <i>Senior Backend Software Engineer & Scrum Master</i>	Beijing, China Feb 2018 – Jul 2021
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- Migrated a Django monolith to Flask/Go microservices on ECS and built a unified connection workflow for 30+ platform integrations with MFA handling for Apple/Google — boosting API performance 50+% and connection success 30+%.

VIPKID <i>Software Engineer</i>	Beijing, China Aug 2016 – Feb 2018
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TALKS & PUBLICATIONS

How Duolingo Built Workflow-as-a-Service with Temporal Nexus <i>Talk, Temporal Replay</i>	2026
Scaling Developer Productivity with Temporal Nexus & Duolingo <i>Webinar</i>	2025
Duolingo simplifies self-service infrastructure with Temporal Nexus <i>Case Study, Temporal.io</i>	2025

SKILLS

Languages: Python, Go, JavaScript, Java, SQL, R
Cloud: AWS (Solutions Architect Professional), EC2, ECS, EKS, IAM; Terraform, Terragrunt, Docker, Kubernetes
Data Storage: PostgreSQL, MySQL, Redis/Valkey, DynamoDB, S3
Workflow & Observability: Temporal, Grafana, Sentry, Honeycomb, Kibana, Datadog, New Relic
AI/LLM: OpenAI Agents SDK, Claude Agent SDK, Temporal-based agentic workflow infrastructure